CS305: Computer Networking

2025 Fall Semester Written Assignment # 1

Please answer questions in English. Using any other language will lead to a zero point.

Q 1. List the five layers of the Internet protocol stack. For each layer, offer a concise explanation using one sentence, and give an example of a protocol or specific networking technology of that layer.

Q 2. For the following pairs of technical terms, provide a definition for each and explain the primary differences between them. Your response should be clear and concise.

(a) "Circuit-switching" and "packet-switching"

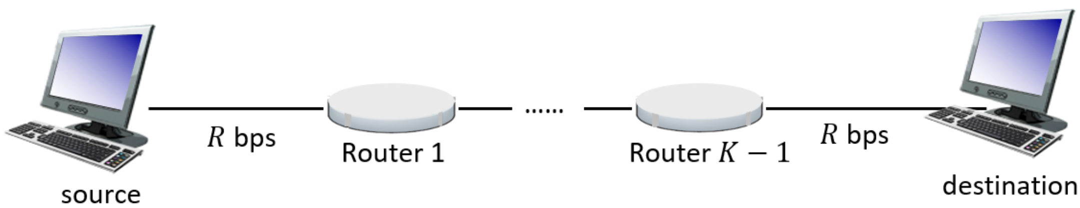
(b) "Client-server" and "peer-to-peer"

Q 3. Answer the following questions:

(a) What type of applications is TCP better suited for? Any application examples?

(b) What type of applications is UDP better suited for? Any application examples?

Q 4. Consider a packet of L bits sending from the source to destination through a K-hop path. That is, there are $K-1$ routers between the source and destination. Suppose each link has a transmission rate of R bits per



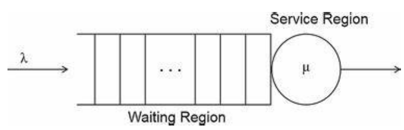
second (bps), and the propagation delay is d for each hop.

(a) Consider a packet switching network. Suppose there is no nodal processing delay and queuing delay. What is the end-to-end delay?

(b) Consider a circuit switching network. Suppose the circuit setup time is τ seconds. Links in the network use frequency division multiplexing (FDM), where the associated frequency band is divided into F subbands, each being allocated to a user (or circuit). [Hint: As a result, the transmission rate allocated to each user (or circuit) is R/F.] What is the end-to-end delay?

(c) Consider a packet switching network with $L = 1000$ bits, $K = 2$, $R = 20$ Mbps, $d = 10\mu s$. There are two packets sent one after the other, and there are no other packet in the system. Let the nodal processing delay at the router be $5\mu s$. We ignore the nodal processing delay at the source and destination. Compute the time required to send both packets from the source and destination. [Note: In our lecture, we set 1 Kbit = 10^3 bits and 1 Mbit = 10^6 bits.]

Q 5. Consider a set of packets with size 10 Mbits and a queue. These packets arrive at the queue with certain patterns defined in (a) and (b), waiting for transmission. The transmission rate is 10 Mbps.



(a) Suppose there is one packet arrival every second. What is the average queuing delay of these packets?

(b) Suppose K packets arrive simultaneously every K seconds. What is the average queuing delay of these packets?

(c) What are the traffic intensity of the scenarios considered in (a) and (b)? Any insights?

Q 6. Consider the following message and answer questions.

```
GET /cs453/index.html HTTP/1.1<cr><lf>Host:
gaia.cs.umass.edu<cr><lf>User-Agent: Mozilla/5.0 (Windows;U; Windows NT
5.1; en-US; rv:1.7.2) Gecko/20040804 Netscape/7.2 (ax)
<cr><lf>Accept:ext/xml, application/xml, application/xhtml+xml, text
/html;q=0.9, text/plain;q=0.8, image/png,*/*;q=0.5<cr><lf>Accept-Language:
en-us, en;q=0.5<cr><lf>Accept-Encoding: zip,deflate<cr><lf>Accept-
Charset: ISO-8859-1, utf-8;q=0.7,*;q=0.7<cr><lf>Keep-Alive:
300<cr><lf>Connection:keep-alive<cr><lf><cr><lf>
```

(a) Does HTTP message run on top of TCP or UDP? Why is TCP or UDP a better choice? Please explain the reason by considering the features of TCP or UDP.

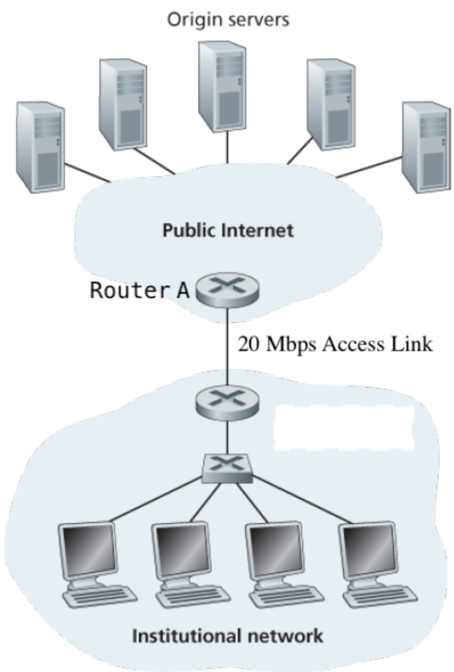
(b) Is this message an HTTP request message or an HTTP response message?

(c) Does this message corresponds to a non-persistent or a persistent connection?

(d) Suppose the browser who sent this message was assigned an identification number 1150 by the associated host. What is the corresponding entry in the cookie file of the browser? To specific its cookie, which header line (including header field name and value) should be included in the message?

(e) If the server receives this message successfully and is going to return the requested object in a message, what would be the status line? What would be included in the entity body?

Q 7. Consider the following figure with an institutional network connected to the Internet. There are a set of objects with size 1 Mbits. Suppose the institution's browsers has an average request rate of 10 requests per second, and all those requests are sent to the origin servers. The average response time is determined as follows:



$$\texttt{Average Response Time} = \texttt{Internet Delay} + \texttt{Average Access Delay}. \quad (1)$$

- Internet delay is the round trip time between router A and the origin server. It is equal to 2 seconds.

- Average access delay is the delay from Router A to the institution router. The transmission rate of the access link is 20 Mbps. The average access delay is equal to $\Delta/(1-\Delta\beta)$, where Δ is the average time required to transmit an object over the access link (i.e., the transmission delay of an obejct); β is the arrival rate (in requests per second) at the access link.

- Note: The delays over other links, e.g., local area network (LAN) delay, are regarded as zero.

Answer the following questions:

(a) Derive average response time of the system.

(b) Suppose there is a cache installed in the institutional LAN. Compute the hit rate $x \in [0,1]$ that leads to an average response time that is less than 1 second.

Q 8. Suppose you click a web page within your Web browser, and your local DNS server has stored the related resource records. Let RTT_0 denote the round trip time (RTT) between your host and your local DNS server. On the web page you visit, there are an HTTP basic file and 12 referenced objects. Let RTT_1 denote the RTT between the local host and the Web server. The HTTP basic file has a size of L Mbit, and each referenced object has a size of L Mbit. The transmission rate is R Mbps. Compute how much time elapses from when you click the link until your web browser receives the objects.

(a) Non-persistent HTTP with no parallel TCP connections?

(b) Non-persistent HTTP with the browser configured for 4 parallel connections?

(c) Persistent HTTP? Note that in this case, if a client knows the URLs of its requested referenced objects, the client can send requests of referenced objects back-to-back without waiting for the responses.

Q 9. Consider distributing a file of F bits to N peers using a client-server architecture. Assume a fluid model where the server can simultaneously transmit to multiple peers, transmitting to each peer at different rates, as long as the combined rate does not exceed u_s.

(a) Suppose that $u_s/N \le d_{min}$. Specify a distribution scheme that has a distribution time of NF/u_s.

(b) Suppose that $u_s/N \ge d_{min}$. Specify a distribution scheme that has a distribution time of F/d_{min}.

(c) Conclude that the minimum distribution time is in general given by $\max\{NF/u_s, F/d_{min}\}$.

Q 10. Explain why a hierarchical DNS system (instead of a huge centralized server) is used.